

The Curved-Corpus Program, Unified: Learned Latent Curves, a Null- Standardized Question Space, and Spherical Defocus, with Machine-Checked Invariants

Shant Tchatalbachian

August 13, 2026

Introduction

The curved-corpus program has produced two papers and a running argument between them. The first (Tchatalbachian, 2026a) built the instrument: a family of learned latent curves on $[0, 1]$ and on the Riemann sphere S^2 , a single-action audit atom with an only-positive- Δ invariant, and a recursive improvement loop that drove three corpora to full primitive coverage across 1391 dispatches with zero regressions. The second (Tchatalbachian, 2026b) audited the instrument: it showed the headline observable $V_2 = PC1 + PC2$ is 98% fixed by marginals, that its celebrated transition was a basis change, that its gate is a rank test a noise corpus passes exactly, retracted the program's most-cited intermediate correction, inverted the Hodge reading (the corpus is anti-cyclic), and replaced the sequence of candidate X 's with a null-standardized question space issuing exclusion-only verdicts.

This paper does three things. First, it **merges** the two accounts into one, keeping the second paper’s corrections as binding on the first’s claims (Sections 2–3). Second, it adds an **optical reading** with exact rather than metaphorical anchors, and uses it to organize a chain the program has been circling: question space $\rightarrow$ intent space $\rightarrow$ null space $\rightarrow$ latent space of 0 through N dimensions (Section 4). Third, it opens a **diffusion channel**: geodesic (slerp) interpolation on the hypersphere generates a Brownian defocus that is diagonal in the program’s own harmonic basis, and experiment G1 — executed here — verifies the governing semigroup identity on the sampling lattice and measures its instructive failure on the real corpus (Section 5). A short section records which of the program’s statements are identities and proves them in Lean 4 (Section 6); scope and caveats close (Section 7).

Two disciplines are inherited verbatim from the 2026-08-12 paper: no statistic is reported without its matched null, and no coordinate is admitted to the map without a demonstrated non-degenerate null. The optical and diffusion vocabulary introduced here is held to the same bar: each term is used only where a metric and a variational problem are both specified, and each proposed coordinate arrives with the null it must pass before admission.

The curve family

Flat curve, ridge, gate

For output dimension $j = 1, \dots, D$ and ordering coordinate $t \in [0, 1]$,

$$z_j(t) = a_{j,0} + \sum_{m=1}^k (a_{j,m} \sin(2\pi f_m t) + b_{j,m} \cos(2\pi f_m t)),$$

with k shared learned frequencies; stacking the design vector $\phi(t)$ gives $\gamma(t) = C\phi(t)$, solved at fixed frequencies by the Tikhonov ridge

$$C^* = \left(\Phi^\top \Phi + \lambda I \right)^{-1} \Phi^\top Z, \quad \lambda = 10^{-3}.$$

The coordinate t is the rank of the top principal component of the z -scored feature matrix, and the fit gate is

$$V_2 = PC1 + PC2 = \frac{\lambda_1 + \lambda_2}{\sum_{a=1}^D \lambda_a} \geq 0.40.$$

Hyperspherical variant, Möbius chart, Fibonacci sampling

For $\mathbf{x} \in S^2$,

$$\gamma(\mathbf{x}) = \mathbf{b} + \sum_{\ell=0}^L \sum_m \mathbf{a}_{\ell,m} Y_{\ell,m}^{S^2}(\varphi_{\theta}(\mathbf{x})), \quad n_{basis} = 16 \text{ at } L = 3,$$

with the Möbius reparameterization $\varphi_{\theta}(z) = (az + b)/(cz + d)$, $ad - bc = +1$, six real degrees of freedom, *frozen at the identity in every run reported to date*. Items reach S^2 by the stereographic lift $\sigma(u, v) = (2u, 2v, u^2 + v^2 - 1) / (1 + u^2 + v^2)$ of their PCA top-2 coordinates. Sampling uses Vogel's golden-angle lattice

$z_i = 1 - \frac{2i+1}{N}$, $\phi_i = 2\pi i/\varphi_g$, $\theta_i = \arccos z_i$, $\varphi_g = \frac{1+\sqrt{5}}{2}$, under which the Gram matrix of the 16 real harmonics is near-orthonormal, $B^{\top} B \approx (N/4\pi) I_{16}$, condition number 1.003–1.006 in every design of this program.

The single-action atom

The smallest audit unit maps a file to one point $p \in S^2$ (coverage vector $\rightarrow$ PCA top-2 $\rightarrow$ lift), defines the ideal pole p^* as the lift of the all-ones vector, the geodesic gap $d(f) = \|p - p^*\|_2$, and selects among single missing-primitive flips the one minimizing post-flip distance.

Lemma 1 (atom invariant). *For any file f and any action selected by the geodesic-only criterion, $\Delta_f = d_{pre} - d_{post} \geq 0$.*

Theorem 1 (linear composition). *For a corpus of N files with atomic actions $\alpha_1, \dots, \alpha_N$, $\Delta_{corpus} = \sum_i \Delta_{f_i}$; if every atomic $\Delta \geq 0$ then $\Delta_{corpus} \geq 0$.*

Corollary 1 (cumulative monotonicity). *The running total over cycles is monotone non-decreasing; the per-cycle increment is unconstrained in shape.*

Empirically: 1391 dispatches over three corpora (skills/ 474, docs/ 126, refs/ 791), 598 strictly positive Δ , zero negative, cumulative $\Delta = +59.7671$, all three corpora absorbing at full coverage; the realized Δ depends only on the missing-primitive count k , unimodal

with peak at $k = 5$ ($\Delta = 0.1206$), identically on all three corpora. The matched-parameter ablation favors the sphere over a capacity-matched flat Fourier baseline (49-item $\delta = +0.977$; 70-item $\delta = +1.342$; single-seed point estimates), and the synthetic-manifold benchmark holds both controls (torus: flat wins 50/50 seeds, $p < 10^{-15}$; sphere: sphere wins 50/50, $p < 10^{-55}$). Section 6 formalizes all three statements above.

The audit, carried forward

The corrections of the 2026-08-12 paper are restated here as binding, in compressed form; every number is from the shipped evidence bundle and its reproduction manifest.

C1 (dimension artifact). V_2 is a trace-normalized share and decays in D at fixed rows (flat-spectrum floor $2/D$); on identical 2286 rows, $V_2 = 0.7235$ at $D = 9$ and 0.2940 at $D = 24$. The celebrated $+0.4664$ “transition” was this proposition evaluated at two points.

C2 (the gate is a rank test). The shipped estimator reports $\hat{r} = 2/V_2$, so $V_2 \geq 0.40 \Leftrightarrow \hat{r} \leq 5$ is a definitional identity (formalized in Section 6); the $(\ell = 384, m = 3)$ variant and 2-D PCA return exactly 1.0000 on all six corpora including single-class iid noise. Exact 1.0000 is a red flag; Section 4 gives it a geometric name.

C3 (corrected null, and a retraction). Under the fixed-margin curveball null (two independently validated uniform samplers), the 2286×9 matrix has null $V_2 = 0.709180 \pm 0.001183$ against real 0.723529 : $\Delta V_2 = +0.0144$, $z = +12.13$, $\mathbb{E}_0[V_2]/V_2 = 0.9802$. The intermediate value 0.8005 ($z = -45.8$) is retracted; its sign was wrong.

C4 (no critical point). $\Delta V_2(N)$ is flat ($+0.0135$ to $+0.0175$) for $N \geq 79$; the previous $N_c \approx 40$ was the null’s own finite-size inflation. P4, P5, P7 of the Ginzburg–Landau scoreboard are falsified or unsupported on execution.

Hodge channel. $\rho_g = 0.99961$, and against a degree-matched null $z(\rho_h) = -53.1$, $z(\beta_1) = -312.5$: the corpus is strongly anti-cyclic; all three pre-registered Hodge predictions are falsified. Only directional

statements are construction-invariant.

The question space. The deliverable is the triple $(\Phi, \mathcal{M}, \mathcal{V})$: a map from corpus to null-standardized statistic vector

$$s(M) = \left[V_2, \Delta V_2 z, PR, Otsu, \bar{\pi}, sd(\pi) \right],$$

reference families generated at the observed $(N, d, \text{density})$, and exclusion-only verdicts at $\max_c |z_{f,c}| > 3$. Round trip: a planted corpus excludes the origin at $z \approx 28$ and recovers its own family at $\max |z| = 0.99$. Placement: yubiOS is M_0 -excluded at $z = +12.3$, nearest a two-population mixture; mean-field Ginzburg–Landau is excluded on the shape of $\Delta V_2(N)$ and, via the dynamics channel (zero backward transitions in 1178 re-measured transitions; no equilibrium ensemble), in principle. The designed counterpart — the curve-compass $\pm$ atom with Metropolis–Hastings acceptance on the measured Φ ladder — restores an exact equilibrium ensemble

$$\pi T(k) \propto \binom{9}{k} e^{-\Phi(k)/T} \quad \text{with detailed balance not rejected at}$$

$\max_k |z(J)| = 0.010$ and an energy-entropy crossover at $T \times = 0.041143$; T is a designer's knob, not a corpus observable.

Calibration. The standard-candle generator plants a spherical-harmonic curve of amplitude s on the lattice of Eq. (5); $\Delta V_2 z$ has power monotone in s over $s \geq 0.5$, power 1.00 at one log-odds by $N = 512$, FPR 8.3% one-sided at nominal 5% (z over-dispersed by roughly a third); the fold-aggregated ladder converts four power-floor cells into one monotone family test ($t_{fold} = 45.19$ paired against a null-ladder $q_{95} = 0.0997$).

The optical reading

Exact anchors

Three identifications are exact, not metaphorical.

(a) *The Möbius chart is a conformal lens.* For an analytic map $w(z)$, geometrical optics assigns the index profile $n(z) = |w'(z)|$: straight rays in the virtual plane map to designed curved rays in the physical plane (Leonhardt 2006; Schmiele et al. 2010). The chart $\varphi_\theta \in PSL(2, \mathbb{C})$ of Eq. (4) is exactly such a map on the Riemann sphere. Frozen at the identity — as in every run of this program to date — it is an unpowered lens ($n \equiv 1$); refining it is lens design in the literal optical sense, with $|\varphi_\theta'|$ the design variable.

(b) *The atom is a discrete Fermat principle.* Rays in an isotropic medium are stationary curves of $\int n ds$, i.e. geodesics of the conformally flat metric $g = n^2 \delta$; Snell's law is the corner condition at an index discontinuity (Kalaba and Ueno 1974; Tyc 1997). The geodesic-only criterion of Lemma 1 is this variational principle discretized: each dispatch takes the step minimizing geodesic distance to the pole, and $\Delta \geq 0$ is the statement that rays never move backward in optical path length. The measured $\Phi(k)$ ladder is the eikonal evaluated on coverage shells, and its telescoping identity (Section 6) is the discrete Hamilton-Jacobi statement.

(c) *Null standardization is deflection against an unlensed background.* Gravitational lensing defines deflection only relative to the path light would have taken without the lens. The map Φ does the same: the fixed-margin ensemble is the medium, and Lyu and Mukherjee (2024) give its spectrum analytically (margin-conditioned matrices converge to a tilted independent-entry ensemble with a variance profile; Marchenko-Pastur at constant margins). The program's finding that 98% of V_2 is marginal-fixed is a property of that medium; $\Delta V_2 = +0.0144$ at $z = +12.3$ is the deflection. Standardization $z_{f,c}$ is a statistical equivalence principle: coordinates always exist in which the medium is locally flat, and what cannot be transformed away — the null-standardized residuals — is curvature.

Caustics

A caustic is a locus where the ray map degenerates: intensity diverges and the image carries no information about the source. The program's exact-1.0000 gate passes are caustics in this sense. All 14 machine-precision $V_2 = 1.0000$ rows of the dimension ladder occur where the sampled design has numerical rank 2; a pure-noise corpus passes

there, so the “focus” is total and totally uninformative. The red-flag rule of C2 (report numerical rank and condition number beside any exact gate pass) is a caustic detector, and the anti-caustic constraint becomes an explicit design rule for any powered lens: reject any φ_θ whose design rank drops or condition number diverges. Classification of these degeneracies (fold vs. structural collapse) is catastrophe-theoretic and left open.

The chain: question space to latent dimensions

The chain the program has been circling is now stated with each arrow’s backing:

- **Question space** $\mathcal{Q}$: the triple $(\Phi, \mathcal{M}, \mathcal{V})$. Corpora are rays; reference families are sources.
- **Null space** $\mathcal{N}_0$: the matched-null ensemble, the vacuum metric and origin of the map. Section 5 upgrades this from a static origin to a dynamical endpoint.
- **Intent space** $\mathcal{I}$: the sufficient-statistic quotient of $\mathcal{Q}$. By Čencov’s theorem the Fisher-Rao metric is (up to scale) the unique geometry invariant under sufficient-statistic morphisms, so the geometry of intent is not a free choice. Constructively, the embedding networks of simulation-based inference in lensing cosmology realize $\mathcal{I}$ as a learned summary space; the admissibility condition of the map applies: $\mathcal{I}$ enters only with a non-degenerate null (train the same embedding on curveball draws; if it recovers the same verdicts, the coordinate is vacuous).
- **Latent space, $0 \rightarrow N$ dimensions**: the effective dimension $D_\lambda = \text{tr}((T + \lambda I)^{-1} T)$ of the kernel integral operator runs continuously from 0 ($\lambda \rightarrow \infty$) to numerical rank ($\lambda \rightarrow 0$); the program’s discrete basis ladder becomes a smooth dial, and the fitted null laws of C4 are the medium’s dispersion relation, dimension playing wavelength. The Parseval per-degree shares E_ℓ are the dimension-invariant (achromatic) coordinates.

Spherical defocus: interpolation, diffusion, and experiment G1

Slerp is the deterministic half of a sphere Langevin step

Geodesic interpolation on S^n is Shoemake's slerp,

$$\text{slerp}(p, q; t) = \frac{\sin((1-t)\Omega)p + \sin(t\Omega)q}{\sin\Omega}, \quad \cos\Omega = p \cdot q,$$

the constant-speed geodesic from p to q . Three objects of the program are discrete slerps already: the atom's flip sequence (a geodesic walk toward the pole, one quantum per move), the fold ladder (constant-ratio steps along a one-parameter family), and the cross-corpus curve alignment. The continuous object they point at is the Riemannian Langevin diffusion

$$dX = -\nabla_g \Phi(X) dt + \sqrt{2T} dB_{S^2}, \quad \pi_T(x) \propto e^{-\Phi(x)/T},$$

whose Euler step is "slerp toward the pole, then tangential noise." The curve-compass $\pm$ atom is the zero-temperature discretization: its $\pi_T(k)$ is the k -shell marginal of Eq. (8) under the exchangeability identity (the potential depends on the coverage count alone), and the $T \rightarrow 0$ limit reproduces the historical absorbing log exactly as measured (0/598 backward transitions). The natural stationary families are von Mises-Fisher, the sphere's Gaussians, with concentration $\kappa \sim 1/T$ as the focus knob; the nearest-family mixture M_4 becomes a two-component vMF mixture on S^2 .

The basis diagonalizes diffusion

The real harmonics satisfy $\Delta_{S^2} Y_{\ell m} = -\ell(\ell+1)Y_{\ell m}$, so Brownian motion (generator Δ , kernel $e^{t\Delta}$) acts diagonally on the program's own basis:

$$\mathbb{E}[Y_{\ell m}(B_t^x)] = e^{-\ell(\ell+1)t} Y_{\ell m}(x), \quad K_t(x \cdot y) = \sum_{\ell} \frac{2\ell+1}{4\pi} e^{-\ell(\ell+1)t} P_{\ell}(x \cdot y).$$

Three consequences. First, on a quasi-uniform design the per-degree Parseval energies of a fitted field decay in closed form, $E_\ell(t) \approx E_\ell(0)e^{-2\ell(\ell+1)t}$: defocusing a corpus is an operation on the already-measured spectrum. Second, a diffusion-time estimator $\hat{t}$ falls out of the per-degree decay — subject, like every coordinate, to the membership condition (its null: $\hat{t}$ on curveball draws). Third, and structurally: Brownian motion on a compact manifold converges to the uniform measure, so **the null space is the $t \rightarrow \infty$ endpoint of the forward heat flow**. “Lensing the question space to the null space” is thereby an exact dynamical statement: forward diffusion is the map $\bar{Q} \rightarrow \mathcal{N}_0$ run continuously, and the reverse-time, score-guided flow of Riemannian diffusion models (De Bortoli et al. 2022; Huang et al. 2022) is the general (non-conformal) lens that focuses the uniform measure back into a corpus distribution, with the learned score $\nabla \log p_t$ as the index gradient. The conformal lens of Section 4 is its six-parameter interpretable subfamily.

Experiment G1: executed

G1 tests Eq. (9) directly (seed 20260813, numpy only; script and JSON record in the session bundle).

Leg 1 (semigroup identity): verified. Fibonacci lattice $N = 2048$, tangent-space Euler–Maruyama, 48 replicates, pooled per-degree regression of $\mathbb{E}[Y(X_t)]$ on $Y(X_0)$:

G1 Leg 1. Measured decay against $e^{-\ell(\ell+1)t}$; ratio measured/predicted $\in [1.000, 1.028]$, the residual growing with $t\ell(\ell+1)$ as first-order Euler discretization predicts.

t	$\ell = 1$ meas./pred.	$\ell = 2$ meas./pred.	$\ell = 3$ meas./pred.
0.01	0.98024 / 0.98020	0.94187 / 0.94176	0.88712 / 0.88692
0.05	0.90522 / 0.90484	0.74160 / 0.74082	0.54962 / 0.54881
0.10	0.82118 / 0.81873	0.55367 / 0.54881	0.30642 / 0.30119
0.20	0.67374 / 0.67032	0.30578 / 0.30119	0.09324 / 0.09072

Leg 2 (real corpus): the closed form does not govern, and the deviation is the finding. The 2286×9 master matrix was run through the paper pipeline (z-score, PCA top-2, RMS rescale,

stereographic lift; ridge on the 16 real harmonics). Per-degree energy shares at $t = 0$: $[E_0, \dots, E_3] = [0.3457, 0.3352, 0.2752, 0.0439]$.

Diffusing the *point positions* (targets fixed; 12 replicates per t) gives:

G1 Leg 2. Point-diffusion energy decay on the real corpus against the closed form.

t	ℓ	measured $E_\ell(t)/E_\ell(0)$	closed form $e^{-2\ell(\ell+1)t}$
0.005	1 / 2 / 3	0.388 / 0.100 / 0.323	0.980 / 0.942 / 0.887
0.020	1 / 2 / 3	0.366 / 0.030 / 0.167	0.923 / 0.787 / 0.619
0.050	1 / 2 / 3	0.395 / 0.018 / 0.085	0.819 / 0.549 / 0.301
0.100	1 / 2 / 3	0.386 / 0.014 / 0.067	0.670 / 0.301 / 0.091

The energy collapses at $t = 0.005$ (RMS geodesic displacement ≈ 0.14 rad) where the closed form permits at most 6% loss; the decay is non-monotone in ℓ ($\ell = 3$ outlives $\ell = 2$ at every t), impossible for heat-kernel decay of a smooth field; and a floor persists ($\ell = 1$ stabilizes near 0.39). The mechanism is the corpus's atomicity: only 176 distinct rows of 512 possible, so the point cloud is a set of heavy point masses, the fitted energy is carried by atomic coincidences rather than smooth harmonic structure, and an infinitesimal diffusion splits each atom into a cloud and kills the coherent fit. Leg 1 pins the identity; Leg 2 measures where its hypothesis (quasi-uniform design) fails. The gap is promoted to a diagnostic,

$$A_\ell(t) = e^{-2\ell(\ell+1)t} - \frac{E_\ell(t)}{E_\ell(0)},$$

large at small t iff the corpus's spherical energy is atomic ($A_1(0.005) \approx 0.59$ on yubiOS). This is the same fact the Hodge channel reached through 176 distinct rows and marginal dominance, now as one scalar; and it gates the generative extension: a Riemannian score-based model of this corpus must smooth the atoms first (vMF bandwidth as pre-diffusion) or it will model point masses. $A_\ell(t)$ is *not yet admitted* to the map: its null (the same quantity on curveball draws, which share the atom structure of the marginals) is specified but not executed.

What the channel adds to the map

(i) A scale-space: $z(t)$ profiles for every coordinate, corpus and null diffused identically, with the ultra-small- t de-atomization window as the informative regime for atomic corpora. (ii) The slerp fold: geodesic rungs $slerp(p_0, p^*; t_k)$ replace the log-odds amplitude ladder, making the fold intrinsic and composable with the achromatic Parseval coordinates. (iii) A generative standard-candle v2 by reverse diffusion, gated on de-atomization. (iv) A single knob unification: the Schrödinger bridge between null and corpus interpolates entropic diffusion ($T > 0$) and optimal transport ($T \rightarrow 0$), the same T the compass swept on the k -marginal.

Machine-checked invariants (Lean 4)

The 2026-08-12 paper drew a boundary between identities (true by construction, carrying no data) and measurements, and showed the program had repeatedly mistaken one for the other. That boundary is now itself machine-checked. The file `CurvedCorpus.lean` (core Lean 4 toolchain, no `mathlib`) states and proves:

1. `atom_delta_nonneg` — Lemma 1 in abstract form: the geodesic-only argmin over candidates including the stay option satisfies $\Delta = d_{pre} - \min(\cdot) \geq 0$.
2. `corpus_sum_nonneg` — Theorem 1: a sum of non-negative per-file deltas is non-negative.
3. `cumulative_monotone` — Corollary 1: appending a non-negative per-cycle increment never decreases the running total.
4. `gate_rank_identity` — C2's equivalence $V^2 \geq 2/5 \Leftrightarrow 2/V^2 \leq 5$ in fraction form ($5p \geq 2q \Leftrightarrow 2q \leq 5p$ for positive p, q): provable by linear arithmetic alone, which is the formal content of “the gate-rank equivalence carries no empirical information.”

5. `phi_ladder_telescope` — D3's identity:
 $\sum_{k < n} (\Phi(k) - \Phi(k + 1)) = \Phi(0) - \Phi(n)$; cumulative Δ telescopes by construction, not by discovery.
6. `heat_exponent_monotone`, `heat_exponent_additive` — $\ell \mapsto \ell(\ell + 1)$ is monotone and the exponent is additive in t : higher degrees always defocus faster (Eq. (9)), and $e^{t\Delta}$ is a semigroup.
7. `mh_flux_symm` — the `min`-symmetry
 $\min \left(\Pi_k q_{kk'}, \Pi_{k'} q_{k'k} \right) = \min \left(\Pi_{k'} q_{k'k}, \Pi_k q_{kk'} \right)$, which is the algebraic core of Metropolis-Hastings detailed balance for the compass chain: both directed fluxes equal the same `min`, so they balance identically.

These are deliberately the *identity-type* statements: nothing empirical is provable, and formalizing the identities is precisely how their non-empirical status is certified. The Monte-Carlo results (Tables 1–2, the calibration sweeps, the null tables) remain measurements with reproduction scripts, not theorems. The formalization uses integers and rationals-as-fractions with a hand-rolled `imin` so that no external mathematics library is required; each statement is closed by `omega`, structural induction, or definitional unfolding.

Scope and honest caveats

Inherited. Every caveat of the 2026-08-12 paper stands: the retraction is part of the record; z is over-dispersed by roughly a third (nothing between $|z| = 2$ and 4 is decisive); Hodge magnitudes are construction-dependent; the sensitivity floor is $s \approx 1$ log-odds at $N \leq 512$; the real-GWTC $z = +67$ certifies parameter redundancy, not astrophysics; azimuthal claims remain unsupported on real data, with the $m = 3$ shares below their nulls for a stated mechanism.

Optical language. The identifications of Section 4 are exact only where stated: conformal maps and index profiles (geometrical optics regime; wave effects out of scope), Fermat/geodesic equivalence (stationarity, not global minimality; caustics excepted), lensing-as-null-standardization (a formal correspondence of measurement conventions, not a physical claim). There is no general theorem that

data exhibits optical caustics; the caustic identification here is confined to the rank-degeneracy cases where the design map provably degenerates. “Refractive index,” “focus,” and “caustic” are not coordinates of the map and do not enter verdicts.

Diffusion channel. Leg 1 verifies the semigroup identity on the sampling lattice; it does not validate any corpus claim. Leg 2’s atomicity diagnostic is specified with its null but the null leg is not executed, so $A_\ell(t)$ is not yet admissible. Diffusion time t , like the compass temperature T , is a knob of designed dynamics, not an observable of the corpus; no estimator of a “corpus diffusion time” is offered, and the $\hat{t}$ coordinate proposed in Section 5 awaits its null. The Langevin extension of the compass is designed dynamics on the measured ladder; it does not reopen any equilibrium reading of the historical log (0/598 is untouched).

Lean formalization. The Lean file certifies identity-type statements only, over exact arithmetic (integers, fraction pairs); it does not formalize real analysis on S^2 , the heat kernel, or any probabilistic statement, and it makes no claim about floating-point behavior of the shipped Python. The abstraction boundary is stated in the file header per theorem.

Single-corpus caveats. The matched-parameter ablation remains single-seed; the second-corpus re-run remains the open item of the predecessor. G1 Leg 2 is one corpus; the atomicity mechanism predicts the same collapse on any corpus with heavy row multiplicity, and that prediction is testable on the CIS/SCAP corpora already in the bundle.

Conclusion

Merged, the program reads as one arc. An instrument was built (curves, atom, loop) whose invariants are real but whose headline statistics were, to first order, properties of its own marginals and bases; an audit rebuilt it around a null-standardized question space with exclusion-only verdicts; and the present paper closes the loop twice over. Downward, into certified algebra: the identity-type claims are now Lean-checked, so the line between what the program *proves* and what it *measures* is mechanical. Outward, into dynamics: the frozen Möbius

chart is recognized as an unpowered conformal lens, the null space as the $t \rightarrow \infty$ endpoint of a heat flow that is diagonal in the program's own basis, and the historical absorbing edit log as the zero-temperature corner of a sphere Langevin whose deterministic half is slerp. The semigroup law is verified where its hypotheses hold; where they fail, the failure is measured, named ($A_l(t)$, atomicity), and put behind the same membership condition as every other coordinate. Focusing a feature space, in this vocabulary, is now a concrete open problem with a concrete guard: power the lens (optimize φ_θ under a null-standardized objective), and never let it reach a caustic.

References

S. Tchatalbachian. Learned latent curves and the hyperspherical-harmonic variant: a closed-loop framework for corpus embedding and recursive skill auditing. Internal paper, papers/learned-latent-curves-2026-08-06.tex, 2026a.

S. Tchatalbachian. Is this X ? A null-standardized question space for corpus curvature, with a Hodge-theoretic channel and a calibrated standard-candle generator. Internal paper, papers/is-this-x-2026-08-12-Final.tex, 2026b.

U. Leonhardt. Optical conformal mapping. *Science* 312:1777–1780, 2006.

M. Schmiele, V. S. Varma, C. Rockstuhl, and F. Lederer. Designing optical elements from isotropic materials by using transformation optics. *Physical Review A* 81:033837, 2010.

R. Kalaba and S. Ueno. Dynamic programming, Fermat's principle and the eikonal equation. *JOSA* 64(3):317, 1974.

T. Tyc. Fermat's principle and the general eikonal equation in a static anisotropic medium. *JOSA A* 14(10):2850, 1997.

C. R. Rao. Information and the accuracy attainable in the estimation of statistical parameters. *Bull. Calcutta Math. Soc.* 37:81–91, 1945.

N. N. Čencov. *Statistical Decision Rules and Optimal Inference*. AMS Translations of Mathematical Monographs 53, 1982.

- B. Efron. Defining the curvature of a statistical problem. *Annals of Statistics* 3(6):1189–1242, 1975.
- V. A. Marchenko and L. A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Mat. Sbornik* 72(4):507–536, 1967.
- H. Lyu and S. Mukherjee. Large random matrices with given margins. arXiv:2407.14942, 2024.
- S. Dasgupta and A. Gupta. An elementary proof of a theorem of Johnson and Lindenstrauss. *Random Structures & Algorithms* 22(1):60–65, 2003.
- T. Zhang. Learning bounds for kernel regression using effective data dimensionality. *Neural Computation* 17(9):2077–2098, 2005.
- K. Shoemake. Animating rotation with quaternion curves. *SIGGRAPH* 19(3):245–254, 1985.
- T. White. Sampling generative networks. arXiv:1609.04468, 2016.
- V. De Bortoli, E. Mathieu, M. Hutchinson, J. Thornton, Y. W. Teh, and A. Doucet. Riemannian score-based generative modelling. *NeurIPS*, 2022.
- C.-W. Huang, M. Aghajohari, J. Bose, P. Panangaden, and A. Courville. Riemannian diffusion models. *NeurIPS*, 2022.
- A. Lou et al. Scaling Riemannian diffusion models. *NeurIPS*, 2023.
- L. Ambrosio, N. Gigli, and G. Savaré. A user’s guide to optimal transport. Lecture notes, 2011.
- G. Strona, D. Nappo, F. Boccacci, S. Fattorini, and J. San-Miguel-Ayanz. A fast and unbiased procedure to randomize ecological binary matrices with fixed row and column totals. *Nature Communications* 5:4114, 2014.
- X. Jiang, L.-H. Lim, Y. Yao, and Y. Ye. Statistical ranking and combinatorial Hodge theory. *Mathematical Programming* 127(1):203–244, 2011.
- H. Vogel. A better way to construct the sunflower head. *Mathematical Biosciences* 44:179–189, 1979.

Reproduction manifest (delta over prior bundles)

New artifacts of this paper (session bundle, seed 20260813):

g1/g1_defocus.py (both G1 legs; numpy only; ~3 min), g1/g1-results.json (full record of Tables 1–2), refs/spherical-defocus-g1-2026-08-13.md (the G1 refs note), lean/CurvedCorpus.lean (the seven machine-checked statements of Section 6; core Lean 4, no mathlib). Input data: data/real/per_row_coverage_v3.json from the shipped is-this-x evidence bundle (2286 rows; column sums as in its Appendix A). Everything else — the curveball samplers and their uniformity proofs, the ladder, the candle, the Hodge decomposition, the compass — is unchanged and lives in the two prior bundles under their own manifests.